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Individual Task 2

This individual assignment combines all topics covered up to and including Week 12. Knit your completed R Markdown file (RMD) to HTML and submit both the RMD and knitted HTML onto Moodle.

Note the following:

- This is a 'Guided Gen AI Use' assignment where you are permitted to use generative AI (GenAI) as a supporting tool. Following University guidelines, if GenAI is used, you must submit a short declaration describing the tools used, purpose of use, extent of use, and outline the steps taken to verify the accuracy of the GenAI output.
- You cannot manually modify the data in software like Excel. Anything related to data preparation and analysis must be completed using R.
- You are expected to utilise the techniques covered in this course, where appropriate, and are not required to go beyond them.
- Maintain a tidy and readable RMD file that runs without any errors. Marks will be deducted if reported findings cannot be replicated from running the submitted script.

Data:

The data for this assignment were extracted from LSEG Workspace and publicly available data provided by the World Bank, CDP, and ICLEI – Local Governments for Sustainability. It consists of two datasets:

- A city-level dataset containing information on geographic and economic conditions, national environmental and social performance metrics, and descriptions or indicators of climate risks and vulnerabilities, collaboration on climate-related projects, and planned climate-related projects. The raw data included non-English survey responses, which were translated into English using functions from the cld3 and polyglor packages.
- A firm-level dataset containing information on firm characteristics, environmental performance and policies, and financial and capital market metrics.

env_projects.csv and ***env_controversy.csv*** include the data to use for this assignment. Refer to later pages for variable names and definitions.

Scenario:

Growing concerns about sustainability challenges, including climate change and biodiversity loss, have increased demand for investments that seek to generate both financial returns and positive environmental and social outcomes. Consequently, sustainable finance has become a mainstream consideration for institutional investors, asset managers, and policymakers over the past two decades. Sustainable infrastructure investments and responsible investments are two activities that fall under sustainable finance.

Sustainable infrastructure investments involve financing infrastructure assets that support economic and social activity while minimising environmental impacts and improving long-term resilience. Investments are often made through public sector tenders, public-private partnerships, or direct ownership structures. Examples include funding renewable energy projects, public transportation systems, and wastewater systems. Responsible investment involves incorporating environmental, social, and governance factors, such as carbon emissions, labour practices, and corporate transparency, into investment decision-making. The objective is to better identify risks and opportunities that may affect long-term investment performance while supporting broader sustainability goals.

You work as an entry-level analyst at an investment firm specialising in sustainable infrastructure. The firm also integrates responsible investing principles across its private market portfolios. The firm has industry expertise in waste management and prioritises infrastructure investments related to this area. It also prefers to partner with funding seekers who have experience collaborating with private sector for-profit organisations. The firm has an established track record of financing public sector projects globally through providing loans and engaging in public-private partnerships and intends to continue supporting such initiatives.

Your line manager has asked you to complete the following analyses and report your findings:

- Identify the most common types of climate hazards affecting communities globally.
- Assess the total number of planned projects and the total value of funding sought by region.
- Produce a ranked list of projects that represent potential collaboration and investment opportunities for the firm. Rank the projects in descending order by the investment costs sought.
- Build a classification model that predicts whether a firm may become involved in an environmental controversy. The model is intended for the initial screening of firms that may pose material environmental risks. Firms flagged as high risk will undergo further expert review, making false positives relatively low cost.

Requirements:

1. Conduct appropriate and comprehensive analyses to provide insights on the most common types of climate hazards affecting communities globally. Briefly summarise your main findings. (25 marks)

2. Return the total number of planned projects and the total value of funding sought by region. (5 marks)

3. Produce a ranked list of projects that represent potential collaboration and investment opportunities for the firm. (10 marks)

4. Build the best model you can for predicting whether a firm may become involved in an environmental controversy and evaluate model performance. (60 marks)

Variable names and definitions:

Dataset: ***env_projects.csv***

Variable name	Description
cityid	Unique city identifier
cityname	Name of the reporting city
cityarea	Jurisdiction where the city is located
cityregion	Geographic region grouping for the jurisdiction where the city is located
area_km	Land area of the jurisdiction boundary in square kilometres
area_natural	Percentage range of the jurisdiction area that is natural or modified terrestrial, freshwater, coastal and/or marine ecosystem
area_population	Current or most recent total population size within the city
econ_gdpt	Annual national gross domestic product (GDP) in current USD
econ_gdtg	Annual national GDP growth rate in percentage
econ_emp	Annual national total unemployment rate as percentage of total labour force
econ_inf	Inflation measured by the annual percentage growth rate of the GDP deflator
econ_inv	Annual national gross fixed capital formation as percentage of GDP
econ_spend	Annual general government final consumption expenditure as percentage of GDP
econ_rd	Annual national expenditure on research and development as percentage of GDP
econ_export	Annual value of goods and services exported as a percentage of GDP
env_ghg	Annual total national greenhouse gas emissions in mega tonnes of carbon dioxide equivalent
soc_water	Pecentage of national population using safely managed drinking water services
soc_sanit	Percentage of national population using safely managed sanitation services
climate_hazard	Description of the impacts on vulnerable populations and sectors from the most significant climate hazards faced by the jurisdiction
collab_civil	An indicator reflecting whether the city has collaborated with civil society organisations on climate-related projects before (1 = Yes)
collab_business	An indicator reflecting whether the city has collaborated with businesses and other private sector for-profit organisations on climate-related projects before (1 = Yes)
collab_public	An indicator reflecting whether the city has collaborated with public sector entities on climate-related projects before (1 = Yes)
project_fundingmodel	Identified financing model for the project
project_totalcost	Total cost of the project in USD
project_investmentcost	Total investment cost needed in USD
project_info	Description of planned climate-related project

Dataset: ***env_controversy.csv***

Variable name	Description
firm_id	Unique firm identifier
firm_name	Name of the company
firm_hq	Country of headquarters
industry	Industry group classification
env_score	An estimated environmental pillar score that measures a company's impact on living and non-living natural systems, reflecting how well a company uses best management practices to avoid environmental risks and capitalise on environmental opportunities to generate long-term shareholder value

env_product	Percentage of green products or services as reported by the company
env_co2s1	Scope 1 direct emissions, measuring tonnes of direct carbon dioxide (CO2) and CO2 equivalent emissions from sources that are owned or controlled by the company
env_co2s2	Scope 2 indirect energy emissions, measuring tonnes of indirect CO2 and CO2 equivalent emissions from consumption of purchased electricity, heat, or steam which occur at the facility where electricity, heat, or steam is generated
env_co2s3	Scope 3 value chain emissions, measuring tonnes of indirect CO2 and CO2 equivalent emissions arising from activities across the value chain, including emissions arising from activities such as employee business travel, waste disposal, outsourced activities, and product use by customers
env_energy	Total direct and indirect energy consumption in gigajoules
env_contro	An indicator reflecting the company's exposure to and involvement in environmental-related controversies based on news data (1 = Involvement in one or more environmental-related controversy, 0 = No involvement)
pol_climate	An indicator reflecting whether the company has a policy or commitment statement that pledges to reduce greenhouse gas emissions or improve greenhouse gas emissions efficiency (1 = Yes, 0 = No)
pol_energy	An indicator reflecting whether the company has a policy or commitment statement that pledges to seek third-party verification of their operational energy consumption data (1 = Yes, 0 = No)
pol_envsup	An indicator reflecting whether the company has a supplier or sourcing policy or commitment statement that addresses environmental issues (1 = Yes, 0 = No)
pol_bio	An indicator reflecting whether the company has a policy or commitment statement that pledges to reduce or avoid impacts on biodiversity (1 = Yes, 0 = No)
pol_socsup	An indicator reflecting whether the company has a supplier or sourcing policy or commitment statement that addresses social issues (1 = Yes, 0 = No)
fin_asset	Total assets reported by the company in USD
fin_rev	Total revenue from business activities reported by the company in USD
fin_cor	Total cost of operating revenue reported by the company in USD
fin_capex	Total capital expenditure reported by the company in USD
fin_roa	Return on assets reported as a percentage, calculated as income before discontinued operations and extraordinary items divided by average total assets
fin_roe	Return on equity reported as a percentage, calculated as income available to common equity excluding extraordinary items divided by average common equity
fin_ebitda	EBITDA margin reported as a percentage, calculated as the earnings before interest, taxes, depreciation and amortisation divided by revenue from business activities
fin_cr	Current ratio, calculated as total current assets divided by total current liabilities
fin_dte	Debt-to-equity ratio, calculated as net debt divided by total shareholders' equity
marketcap	Market capitalisation of the company in USD, calculated as the product of the number of securities issued and share price at the end of the financial reporting period